Detecting and Interpreting Text Generated By Large Language Models: A Review of the Literature

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Abstract

Large language models such as ChatGPT have become widely adopted and are increasingly capable of generating text that closely resembles human writing. This has made the automatic detection of machine-generated text (MGT) an important research problem, with direct relevance to academic integrity, content moderation, and misinformation detection. Existing detection approaches span three principal paradigms: supervised fine-tuned classifiers, statistical zero-shot methods, and watermarking. Research in this space has focused heavily on improving detection accuracy, with comparatively little attention paid to interpretability. Current detectors produce classification decisions without explaining which features of a text triggered them, limiting their usefulness in contexts where reasoning must be verified or failure modes diagnosed.

This review examines existing approaches to MGT detection with a focus on interpretability, and considers how related fields such as authorship attribution, stylometry, and explainable AI can inform more transparent and reliable detection methods.

1 Introduction

Large language models (LLMs) are a class of artificial intelligence systems trained on large amounts of text data to predict and generate sequences of words. Given a sequence of words, an LLM estimates the probability of each possible next word and selects from the most likely candidates, repeating this process token by token until a full passage of text is generated. In this paper we will refer to this kind of text as LLM-generated text. Systems such as ChatGPT, GPT-4, and Claude have made this capability widely accessible, enabling users to generate fluent, coherent text across many domains with minimal effort.

Because LLM-generated text can closely resemble human writing, it has become difficult to distinguish machine-generated content from text written by a person. This has raised concerns across several areas: students submitting AI-generated work as their own, the mass production of fake reviews and misinformation, and a broader erosion of trust in written content [23]. These concerns have motivated research into automated methods for detecting machine-generated text (MGT).

From the existing literature, three principal detection approaches have emerged: supervised, statistical, and watermarking. Supervised detectors are classifiers trained on labelled datasets of human-written and LLM-generated text. They use pre-trained language models such as RoBERTa [11], a transformer-based model that learns rich text representations and can be fine-tuned for classification, as a backbone. They achieve strong performance when tested on text similar to their training data, but struggle when applied to new domains or different LLMs. Statistical and zero-shot methods exploit properties of LLM-generated text such as token probability distributions and predictability patterns, without requiring labelled training data. A popular example is DetectGPT [14], which detects generated text by measuring how the probability of a text changes when small modifications are applied to it. Watermarking methods embed hidden statistical signals into text at the point of generation, much like an artist signing a painting, so that it can later be identified as machine-generated [10]. Substantial research has gone into enhancing the accuracy and performance of these approaches, but comparatively little attention has been paid to explaining why a text was classified as machine-generated.

The importance of interpretability in classification systems is well established. Rudin [18] argues that black box models used for high-stakes decisions are problematic precisely because their reasoning cannot be scrutinised, and that interpretable models should be preferred. Cesarini et al. [3] similarly identify the trade-off between accuracy and interpretability as a fundamental tension in machine learning. The history of text analysis offers a concrete illustration of what happens when this is ignored: the CUSUM technique gained widespread attention and was accepted as expert evidence in courts, yet the research community rejected it as unreliable because its reasoning could not be verified [20]. MGT detection faces the same question: how verifiable can a detection decision be if the reasoning behind it cannot be examined? Beyond verifiability, interpretability is necessary for diagnosing why detectors fail in new domains, and for building the kind of user trust required in high-stakes contexts such as academic integrity and content moderation. A probability score alone is not sufficient for an educator or moderator who must justify and act on a detection decision.

This review surveys approaches to MGT detection with a focus on interpretability. Section 2 examines the three principal detection paradigms and their limitations. Section 3 situates the problem within the broader NLP traditions of authorship attribution, stylometry, and explainable AI. Section 4 surveys interpretable approaches to MGT detection specifically. Section 5 discusses the tension between detection performance and interpretability across the literature, and Section 6 concludes with the main takeaways and direction for future work.

2 Background: Detection Methods for LLM-Generated Text

We briefly survey the three principal paradigms for MGT detection. The purpose of this section is to contextualise the subsequent discussion. Each paradigm is covered only to the depth necessary to establish its interpretability properties and limitations.

2.1 Supervised and Fine-Tuned Detectors

Supervised detection methods represent the most widely adopted approach to MGT detection. These methods frame the problem as a binary classification task in which a model is trained on labelled datasets of human-written and LLM-generated text, learning to distinguish between the two classes from examples. The dominant approach is to take a pre-trained language model and fine-tune it for the detection task. Fine-tuning refers to taking a model already trained on large amounts of general text and continuing to train it on a smaller, task-specific dataset. Encoder-based models such as RoBERTa [11] have become the standard backbone for such classifiers, with multilingual variants such as XLM-RoBERTa [4] and DeBERTa [9] extending this approach to cross-lingual settings. A wide range of related transformer-based models have also been explored for this task [23].

Performance in MGT detection is typically evaluated using precision, recall, F1 score, and AUROC. Precision measures the proportion of texts identified as machine-generated that are actually correct, while recall measures the proportion of machine-generated texts that are successfully detected:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{1}$$

The F1 score combines both into a single measure:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN} \tag{2}$$

AUROC, or Area Under the Receiver Operating Characteristic curve, evaluates a model across all possible decision thresholds rather than at a single fixed point, making it the primary metric in MGT detection where threshold sensitivity matters [23].

Fine-tuned detectors achieve strong in-domain performance under these metrics, with accuracy rates of up to 95% when evaluated on text from the same domain and generator as the training data [23]. This performance does not hold beyond training conditions however. MULTITuDE [13], a large-scale multilingual and multi-LLM benchmark, shows that models trained on English achieve a macro F1 score of approximately 0.93 in-domain, dropping to approximately 0.69 on unseen languages, a degradation of over 25%. DetectRL [24] further shows that performance degrades under distributional shift, where differences between training and test data lead to reduced detection accuracy. Sarvazyan et al. [19] attribute this to a reliance on spurious correlations in training data rather than robust linguistic signals.

As black-box models, supervised classifiers provide no insight into the linguistic or stylistic features driving their predictions. While F1 and AUROC quantify how well a model performs, they say nothing about how it makes its decisions. Wu et al. [23] observe that mainstream detection benchmarks evaluate these systems almost exclusively on such metrics, with interpretability unaddressed as an evaluation criterion.

2.2 Statistical and Zero-Shot Detection

Statistical detection methods rely on the probabilistic structure of language models to distinguish machine-generated from human-written text, without requiring labelled training data. A language model assigns probabilities to sequences of text. For a sequence $W = (w_1, w_2, \ldots, w_n)$, it estimates the probability of each token given all preceding tokens, factorising the joint distribution as follows [15]:

$$P(W) = \prod_{t=1}^{n} P(w_t \mid w_1, \ldots, w_{t-1}) \tag{3}$$

Models operate in log space for numerical stability, giving the log-likelihood of a sequence:

$$\log P(W) = \sum_{t=1}^{n} \log P(w_t \mid w_1, \ldots, w_{t-1}) \tag{4}$$

This measures how probable a sequence is under a given model [15]. Zero-shot detection refers to identifying machine-generated text without any task-specific training, relying solely on these statistical properties. Wu et al. [23] categorise statistical detection methods into three groups based on how much access they require to the generating model: linguistic feature statistics, white-box statistics, and black-box statistics.

White-box statistical methods use internal model outputs directly. Logits are the raw unnormalised scores a language model produces before converting them to probabilities, and white-box methods use these to compute detection signals. Two signals derived from the log-likelihood are particularly common. Perplexity measures how uncertain a model is when predicting a sequence [22]:

$$PPL(W) = \exp\left(-\frac{1}{n}\sum_{t=1}^{n} \log P(w_t \mid w_1, \ldots, w_{t-1})\right) \tag{5}$$

Lower perplexity means the model finds the sequence more predictable. Since LLMs tend to favour high-probability tokens during generation, their outputs often exhibit lower perplexity than human-written text under the same model. Token rank is a complementary signal, representing the position of each token in the model's sorted probability distribution at each step.

Building on these signals, GLTR [6] visualises token-level ranks and probabilities through a colour-coded overlay, allowing users to identify patterns of high-probability token usage. It improved human detection accuracy from 54% to 72%. Perplexity alone is insufficient as a detection signal however, as LLM-generated text can exhibit high perplexity depending on the prompt [23].

DetectGPT [14] takes a more sophisticated approach, based on the observation that machine-generated text tends to lie near local maxima of the model's log probability function. It generates slightly modified versions of a candidate text and computes the perturbation discrepancy:

$$d(x, p_\theta, q) \triangleq \log p_\theta(x) - \mathbb{E}_{\tilde{x} \sim q(\cdot|x)}[\log p_\theta(\tilde{x})] \tag{6}$$

where x is the input text, p_θ is the language model, and $q(\cdot|x)$ generates perturbed versions of x with similar meaning. A consistently positive discrepancy indicates the original text sits in a high-probability region, which DetectGPT treats as evidence of machine generation [14]. The method is computationally expensive and vulnerable to paraphrasing attacks, where a text is rewritten to change its surface form while preserving its meaning.

Black-box statistical methods do not require access to model internals. They compare original texts with LLM-generated rewrites,

based on the observation that language models make fewer modifications to their own outputs than to human-written text. These methods show stronger generalisation than white-box approaches but require repeated queries to external models, introducing substantial computational overhead [23].

Linguistic feature statistics take a different approach entirely, analysing surface-level properties of text without any model access. Features such as n-gram frequencies, function word usage, and syntactic patterns have been shown to differ between human and machine-generated text, where an n-gram is a contiguous sequence of n tokens. Hamed and Wu [8] demonstrated that bigram distributions differ substantially between ChatGPT-generated and human-authored academic writing. These approaches are straightforward to apply but depend on large and diverse datasets, limiting their robustness across domains [23].

Statistical methods generally do not require labelled data and are less dependent on specific model deployments than supervised classifiers. Their shared limitation is that performance degrades under adversarial conditions such as paraphrasing, and their outputs, while grounded in model probabilities, remain difficult to interpret for non-technical users.

2.3 Watermarking

Watermarking takes a different approach to MGT detection from the methods discussed so far. Rather than analysing text after generation, watermarking embeds detectable statistical signals during the generation process itself. Wu et al. [23] categorise these methods into three types: data-driven watermarking, which embeds patterns into training data; model-driven watermarking, which modifies the token generation process during inference, that is, when the model is actively producing text; and post-processing watermarking, which embeds signals into text after it has been generated.

The most studied model-driven approach is the logit-based watermarking framework of Kirchenbauer et al. [10]. At each generation step, the vocabulary is partitioned into a green list and a red list using a hash of the preceding token. The model's logit distribution is then biased to favour green list tokens during generation. Detecting the watermark involves checking whether the proportion of green tokens in a text is statistically higher than expected by chance. This operates with minimal impact on text quality and achieves a low false-positive rate.

Watermarking methods share a practical limitation however. They require cooperation from LLM providers at the point of generation, making them inapplicable to text from models where watermarking has not been implemented. Open-source models and third-party deployments present a particular challenge in this regard [23]. Watermarks can also be disrupted through paraphrasing attacks, which alter the surface form of text enough to break the statistical signal. As a detection mechanism, watermarking is therefore best suited to controlled deployment settings rather than general-purpose detection of LLM-generated text in the wild.

A comparative summary of these detection paradigms, including their detection signals, strengths, and interpretability, is provided in Appendix A.

3 Interpretable Text Classification

Having outlined how detection is currently approached, we now examine how interpretability is treated in text classification more broadly.

3.1 Interpretability in NLP: Concepts and Scope

The ability to explain a model's predictions is increasingly recognised as being as important as predictive accuracy, particularly in high-stakes applications [3]. Achieving peak accuracy on modern datasets frequently requires complex models such as ensemble methods or deep neural networks, whose internal reasoning can be difficult to interpret. This tension between accuracy and interpretability has motivated the development of methods aimed at making model predictions more comprehensible to users.

A key distinction is between interpretability and explainability. Although the terms are often used interchangeably, Resck et al. [16] treat them as complementary perspectives. Interpretability refers to understanding the inner workings and internal behaviour of a model, that is, how it processes inputs and arrives at decisions. Explainability focuses on justifying final model decisions by clarifying why a model produced a specific output. Resck et al. [16] identify this as an unresolved problem in current research, noting that bridging this gap remains an open research problem.

A further distinction is between ante-hoc and post-hoc approaches to model transparency. Cesarini et al. [3] define ante-hoc approaches as those that employ the same model for both prediction and explanation, such as linear models whose feature weights directly reveal the factors influencing predictions. Post-hoc approaches generate predictions using one model and explanations using a separate one. This enables them to produce explanations while keeping the prediction model untouched. This distinction is central to the present work. While the majority of Explainable Artificial Intelligence (XAI) research for text classification focuses on post-hoc methods applied to black-box models, of interest to the present work are ante-hoc approaches, where interpretability is built into the model design rather than applied after the fact.

3.2 Feature-Based Approaches: Authorship Attribution and Stylometry

Handcrafted linguistic features have a long history in text classification, predating modern machine learning. Authorship attribution, the task of identifying the author of a text from stylistic and linguistic evidence, is one of the earlier and more developed applications of this approach. Stamatatos [20] traces this tradition back to the nineteenth century, beginning with Mendenhall's quantitative analysis of Shakespeare's plays and the study by Mosteller and Wallace, who applied Bayesian statistical analysis to the frequencies of common function words to resolve the disputed authorship of the Federalist Papers.

Central to this tradition is stylometry, the computational measurement of writing style through style markers. Stamatatos [20] organises these markers into five categories: lexical, character, syntactic, semantic, and application-specific features. Table 1 reproduces the original taxonomy with the computational tools required for each feature type. Lexical features cover token-based measures such as word length, sentence length, vocabulary richness, and word n-grams.

Function words, articles, prepositions, and pronouns, are particularly useful because authors use them unconsciously and they carry little topic-specific information, making them reliable stylistic indicators. Character n-grams are said to be among the most effective character-level features for capturing stylistic nuance while remaining tolerant to noise. Syntactic features capture structural patterns in sentence construction through part-of-speech frequencies, phrase structure, and rewrite rule frequencies. Semantic features exploit higher-level information such as synonym usage and semantic dependencies, though these are harder to measure accurately due to noise from NLP tools.

Table 1: Stylometric feature categories reproduced from Stamatatos [20].

Category	Feature Type	Tools Required
Lexical	Word/sentence length	Tokenizer
	Vocabulary richness	Tokenizer
	Word frequencies	Tokenizer, [Stemmer]
	Word n-grams	Tokenizer
	Spelling errors	Spell checker
Character	Character types	Character dictionary
	Character n-grams	Feature selector
	Compression	Compression tool
Syntactic	POS tags	POS tagger
	Phrase structure	Partial parser
	Rewrite rules	Full parser
	Syntactic errors	Syntactic checker
Semantic	Synonyms	Thesaurus
	Dependencies	Semantic parser
	Functional	Specialised dicts
App-specific	Structural	HTML parser
	Content/language	Specialised dicts

The classifiers used in this field are often interpretable by nature. Profile-based methods use probabilistic models such as Naive Bayes, while instance-based approaches include support vector machines, decision trees, and nearest-neighbour algorithms [20]. This approach of pairing handcrafted features with transparent classifiers has been applied across a range of NLP tasks beyond authorship attribution. Aich et al. [1] apply linguistic features including punctuation ratios, word length, and sentence structure to fake news detection, demonstrating that surface-level stylistic signals carry meaningful discriminative power. Fagni et al. [5] similarly show that stylometric features can distinguish deepfake tweets from authentic ones, and extends this approach to short-form social media text.

Stamatatos [20] identifies a persistent limitation in this area. While researchers can estimate the statistical significance of certain features, producing a human-understandable description of authorial style remains difficult. As later sections discuss, the same limitation applies to MGT detection, where the absence of interpretable models leaves users unable to understand or verify detection decisions.

3.3 Ante-Hoc vs Post-Hoc Interpretability

To understand how interpretability can be incorporated into MGT detection systems, we explore two broad approaches in Explainable Artificial Intelligence (XAI): ante-hoc and post-hoc interpretability.

Ante-hoc methods are inherently interpretable, using models whose internal structure directly reveals how predictions are made. Examples include linear models and decision trees, where feature weights or decision rules can be inspected to understand model behaviour [3]. In contrast, post-hoc methods operate on top of already trained models, generating explanations for predictions without altering the underlying model.

Post-hoc approaches are widely used in modern NLP, where high-performing models such as deep neural networks are typically opaque. These include model-agnostic techniques such as LIME [17] and SHAP [12], which estimate the contribution of input features to a prediction. Explanations can be produced at different levels, including local explanations that justify individual predictions and global explanations that describe overall model behaviour [3, 16].

From the evaluation presented by Cesarini et al. [3], we observe that different post-hoc methods involve clear tradeoffs. Feature-based explanations tend to more closely reflect the behaviour of the underlying model, while rule-based explanations are often easier for users to interpret. Global explanations are generally perceived as more trustworthy, whereas local explanations provide more precise insight into individual decisions. However, these methods can become computationally expensive when applied to text data. For example, SHAP requires substantial memory when used with large feature sets, making it difficult to scale in practice. These findings suggest that no single post-hoc method satisfies all requirements, with tradeoffs between fidelity, usability, computational cost, and interpretability.

Another critique is offered by Rudin [18], who argues that post-hoc explanations of black-box models are fundamentally limited because they cannot be fully faithful to the model being explained. If an explanation were completely faithful, it would be equivalent to the model itself. This raises concerns about the reliability of post-hoc explanations, particularly in high-stakes settings where decisions must be justified.

From the literature considered in this review, more work has been done on explaining black-box systems after training, with comparatively less attention given to models that are interpretable by design. This leaves scope for further exploration of ante-hoc approaches to text classification.

4 Interpretable Detection of LLM-Generated Text

In the context of MGT detection, interpretability refers to the ability to identify which linguistic or stylistic features of a text led to its classification as machine-generated.

4.1 Early Interpretable Detectors: GLTR

GLTR (Giant Language Model Test Room), introduced by Gehrmann et al. [6], is a good example of early methods of making MGT detection explainable to non-expert users. The tool is motivated by a straightforward observation: because LLMs generate text by repeatedly selecting high-probability tokens, their outputs tend to use words that rank highly in the model's predicted distribution at each position. Human writing, by contrast, draws from a wider and more variable subset of the distribution. In essence, this means humans are less likely to choose the highest-probability words and tend to have a more variable vocabulary. GLTR exploits this difference through

three statistical tests applied to each token in a text. Formally, these are defined as:

$$\text{Test 1:} \quad \hat{p}(X_i) = p_{\text{det}}(X_i = \hat{X}_i \mid X_{1:i-1}) \tag{7}$$

$$\text{Test 2:} \quad \text{rank}(X_i) = \text{rank in } p_{\text{det}}(X_i \mid X_{1:i-1}) \tag{8}$$

$$\text{Test 3:} \quad H(X_i) = -\sum_{w} p_{\text{det}}(X_i = w \mid X_{1:i-1}) \log p_{\text{det}}(X_i = w \mid X_{1:i-1}) \tag{9}$$

Test 1 measures how probable the actual word was under the detection model. Test 2 measures its absolute rank in the predicted distribution. Test 3 measures the entropy of the predicted distribution at that position, that is, how uncertain the model was about what word would come next. These tests are visualised through a colour-coded overlay: tokens ranking in the top 10 predictions are highlighted green, top 100 yellow, top 1,000 red, and beyond that purple. Figure 1 shows this overlay applied to both human-written and generated text. The generated sample is dominated by green and yellow tokens, while the human-written text shows a much higher proportion of red and purple tokens, reflecting greater lexical variability.

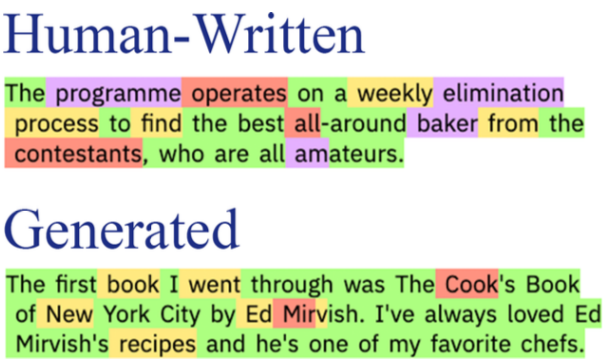


Figure 1: The GLTR top-k overlay applied to human-written and generated text. Green tokens rank in the top 10 predictions, yellow in the top 100, red in the top 1,000, and purple beyond that. The generated text is dominated by green tokens while the human-written text shows greater variation. Reproduced from Gehrmann et al. [6].

Evaluated on 50 articles from three generated and three human sources, GLTR's top-k bucket classifier achieves an AUC of 0.87 using GPT-2 as the detection model, outperforming a bag-of-words baseline [6]. A human-subjects study with 35 participants found that unaided detection accuracy was 54.2%, barely above chance. With the GLTR overlay, accuracy improved to 72.3%, a gain of 18.1 percentage points.

Gehrmann et al. [6] also note that models rarely generate synonyms or alternative referring expressions for the same entity, whereas human writers naturally vary how they refer to things throughout a text. This is consistent with the theory of centering in discourse analysis [7], which describes how humans maintain coherence by varying references. In other words, the lack of referential variety in generated text is itself a detectable stylistic signal, pointing toward discourse-level features as useful indicators of machine generation beyond simple token probability.

4.2 Feature Engineering for MGT Detection

Several studies have examined what makes LLM-generated text different from human writing at a surface level. Wu et al. [23] document that LLM text tends to repeat specific phrases more often, use fewer adverbs, and favour nouns and determiners, giving it a more formal and impersonal tone than human writing. Human text, by contrast, is more varied and less predictable in its word choices.

André et al. [2] tested this directly by comparing AI-generated versions of 2,100 research paper abstracts against the originals. Human abstracts had higher perplexity, more grammatical variation, and more diverse n-gram distributions. A Random Forest classifier trained on these features achieved a precision of 0.986, with perplexity, grammatical error rate, and n-gram distribution ranking as the most discriminative signals. This is consistent with the observation in Section 4.1 that humans draw from a wider part of the probability distribution than LLMs.

At the word level, Hamed and Wu [8] found that bigram distributions alone were enough to correctly identify 98% of LLM-authored academic papers. Aich et al. [1] report similar results for fake news detection using punctuation ratios, word length, and sentence structure, and Fagni et al. [5] extend this to short-form social media content.

Across these studies, the features used are surface-level and readable, the classifiers are transparent, and the decisions can be explained without access to model internals. This stands in contrast to the statistical and supervised approaches covered in Section 2, where detection signals are derived from model probabilities and are difficult to communicate to non-technical users.

4.3 Limitations of Current Interpretable Approaches and the Research Gap

More can be done in the area of interpretable MGT detection. GLTR [6] was built on the assumption that LLMs sample from the head of their probability distribution, which held for GPT-2. Wu et al. [23] note that modern systems trained with instruction tuning and reinforcement learning from human feedback produce far more varied outputs, and whether GLTR's tests remain effective against systems like GPT-4 or Claude is yet to be established. On top of this, GLTR is vulnerable to paraphrasing attacks, where rewriting a text shifts its tokens away from high-probability regions, quietly dismantling the statistical signal the tool relies on [23].

A similar story unfolds with feature-based approaches. Surface-level features such as punctuation patterns and n-gram distributions change when a text is paraphrased, even when its meaning stays the same [24]. This leads us to question how robust these signals truly are in adversarial settings. André et al. [2] demonstrate that surface-level features can be highly effective under normal conditions, yet their application to MGT detection with interpretability as an explicit goal remains uncommon. Detection research has, by and large, favoured methods that rely on model-derived probabilities, which require access to model internals and offer little transparency to the people who actually use these tools [23].

What is perhaps most telling is how the field chooses to evaluate itself. Uchendu et al. [21], in one of the earliest efforts to benchmark human versus machine-generated text detection, measured performance across multiple generators with no interpretability measures in sight. More recent benchmarks follow the same path. MULTITuDE [13] and DetectRL [24] both report accuracy and AUROC as their primary metrics, with no consideration of whether detection decisions can be explained or verified. The field has consistently asked how well detectors perform, rarely stopping to ask whether anyone can understand why. This has left interpretability and explainability largely unaddressed.

5 Discussion and Conclusion

This review set out to answer a straightforward question: can we detect LLM-generated text in a way that is both accurate and interpretable? To get there, we covered a range of approaches and ideas across the literature.

We started with the three main detection approaches. Supervised classifiers perform well, but mainly within the domain they were trained on, and because they operate as black boxes, there is no visibility into what they actually learned [13, 24]. Statistical methods avoid the need for labelled data, but their outputs are difficult to communicate and degrade under paraphrasing and distribution shift [14]. Watermarking is appealing in theory, but depends on cooperation from model providers, which makes it difficult to enforce in practice [10]. Across all three, two issues keep coming up. They are susceptible to adversarial conditions, and they are not designed with interpretability in mind.

To understand what interpretability would look like in this setting, we looked at work in interpretable text classification and explainable AI. This gave us a clearer picture of how features of text, such as lexical, syntactic, and stylistic patterns, can be used not just to make predictions, but to understand them [1, 20]. We also drew a distinction between explainability and interpretability, which matters more than it might first seem. Explaining a decision after the fact is not the same as building a model whose reasoning is transparent from the start [3, 16].

We then looked at authorship attribution, which in many ways can be seen as an earlier form of this problem. It shows that the idea of detecting patterns in writing style is not new, but also highlights a long-standing limitation. Predictions can be made, but are not always easy to justify in a way that users can trust [20].

More recent work such as GLTR shows that interpretability in detection is possible. It provides visual and human-readable signals, and improves how well people can identify machine-generated text [6]. At the same time, it was developed for earlier models, and whether such approaches remain effective for modern LLMs is still an open question. Feature-based methods further show that surface-level linguistic patterns carry useful signal for this task, and can support more transparent decision-making [1, 8].

Putting it all together, the literature shows strong progress on detection performance, but much less attention to interpretability. Most work either focuses on improving accuracy using opaque models, or on explaining those models after the fact using methods such as LIME [17] and SHAP [12]. From the works considered in this review, there has been less emphasis on approaches that are interpretable by design. This leaves space to further explore methods that combine detection with transparent reasoning, particularly for modern LLM-generated text where both are required.

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A Comparison of Detection Paradigms

Table 2 summarises the key differences between these detection paradigms, highlighting the trade-offs between detection signals, performance, and interpretability.

Table 2: Comparison of machine-generated text detection paradigms in terms of detection signals, strengths, and interpretability.

Paradigm	Subtype	Representative Work	Detection Signal	Strengths	Interpretability
Supervised	Fine-tuned classifiers	MULTITuDE [13], DetectRL [24]	Learned contextual representations from labelled data	High in-domain accuracy; strong benchmark performance	Low (black-box)
Statistical	Linguistic features	Hamed and Wu [8]	Stylometric patterns (e.g. n-grams, structure, function words)	Simple; no training required; directly tied to text properties	High (feature-based)
Statistical	White-box methods	DetectGPT [14], GLTR [6]	Token probabilities, rank, entropy, probability curvature	Model-agnostic; theoretically grounded; no labelled data required	Medium
Statistical	Black-box methods	Wu et al. [23]	Similarity between original and LLM-rewritten text	Strong generalisation across domains and models	Low–Medium
Watermarking	Model-driven watermarking	Kirchenbauer et al. [10]	Embedded statistical signals during generation	Reliable detection when watermark is present; low false positives	Medium